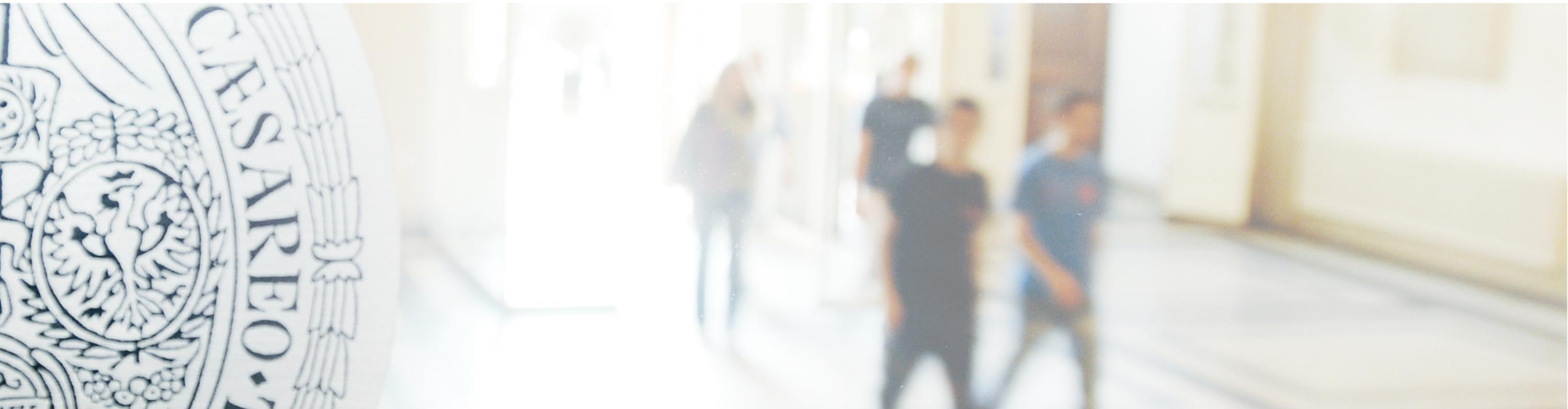


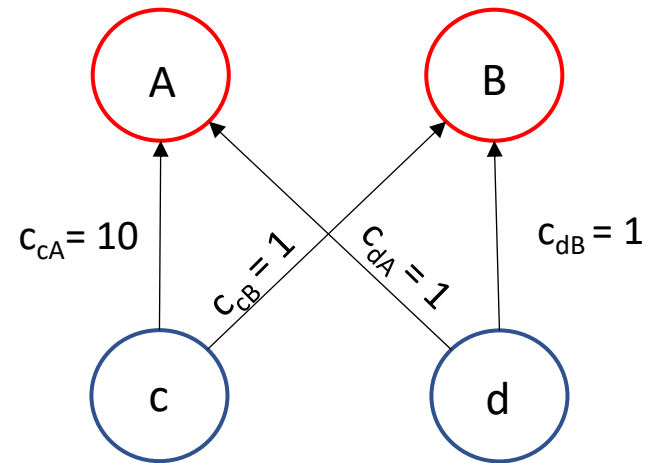


Final Bachelor Thesis Presentation

Finding optimal solutions to a Teacher Assignment Problem using MILP- and SMT-Solvers

Supervisor: Manuel Eberl

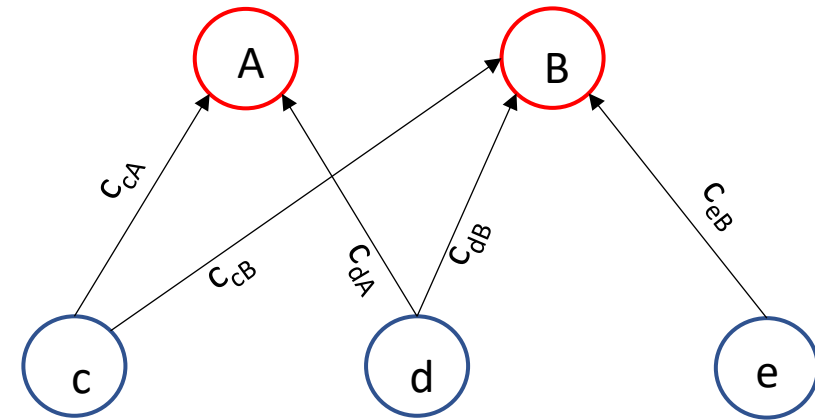
The classical assignment problem (example)



What's special about Teacher Assignment Problems?

Typical Characteristics:

- » More courses than teachers
- » Teachers can get assigned to several courses
- » Teachers are not available for every course
- » Courses can take place simultaneously
- » Teachers have a desired workload



Input

<i>Courses</i>					<i>Available teachers</i>			
ID	Place	Length	Needed	Overlaps	Pref. 0	Pref. 1	Pref. 2	Pref. 3
0	AT-6020	2,5	2	6	Olivia, Emma	Charlotte	Lukas	
1	AT-6020	2,5	2			Jonas		
2	DE-83645	2,5	2		Joel	Jonas, Olivia		
3	DE-93453	8	3	9	Colt			Griffin
4	DE-93453	3,5	2		Jonas		Lukas, Emma	

Additionally:

- » Coordinates of courses and instructors
- » The number of days each instructor would like to work

Output

Course-ID	Assigned teachers			
0	Riggs			
1	Luciana	Meadow	Lane	Morgan
2	XXX			
3	XXX	XXX	XXX	XXX
4	Riggs	XXX		
5	Nicholas	Meadow	Lane	XXX
6	Jaylin	Meadow	Lane	Morgan
7	Jaylin	Meadow	Lane	XXX
8	XXX	XXX		
9	Jaylin	Nicholas	Meadow	Morgan
10	Jaylin	Nicholas	Morgan	
11	Phoebe	Luciana	Meadow	Riggs
12	Dante	Jaylin	Meadow	Lane
13	Nicholas	Lane	XXX	XXX

Complexity of brute force algorithms

With 20 teachers and 40 courses, each requiring two teachers, there is over a googol of combinations

10,000,000,000,000,000,000,000,000,000,000,000,000,000,000,000,000,
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Solution

1. Describe the problem as a Linear Programming Problem
 - Linear cost function
 - Linear constraints
2. Use a solver to find the optimum
 - MILP-solver PuLP
 - SMT-solver Z3 with its vZ extension

Constraints

- i. Teachers cannot teach courses which overlap in time
- ii. Teachers who have priority 3 towards a course must be assigned to the course
- iii. The number of teachers assigned to a course must not exceed the number of teachers needed for the course

$$\sum_{c_i \in s} x_{tc_i} \leq 1$$

$$\forall t \in [1, \dots, T] \quad \forall s \in S$$

$$P_{tc} = 3 \Rightarrow x_{tc} = 1$$

$$\forall t \in [1, \dots, T] \quad \forall c \in [1, \dots, C]$$

$$\sum_{t=1}^T x_{tc} \leq R_c$$

$$\forall c \in [1, \dots, C]$$

Criteria of the cost function

- i. The overall distance that needs to be travelled
- ii. The priorities the teachers have towards their assigned courses
- iii. How well the desired workloads of the teachers were met

Easy! Minimize the sum.

Easy! Maximize the sum.

Hmm... Minimize the standard deviation?

Teachers	Desired workdays	Assigned workdays	Absolute difference	Relative difference
Max	2	10	8	500%
Sarah	10	10	0	0%
Peter	10	20	10	100%

Linearization of the standard deviation

Instead of minimization of the standard deviation:

- » Minimize the average relative deviation of desired to assigned workload
- » Minimize the maximum relative deviation of desired to assigned workload

Weighting:

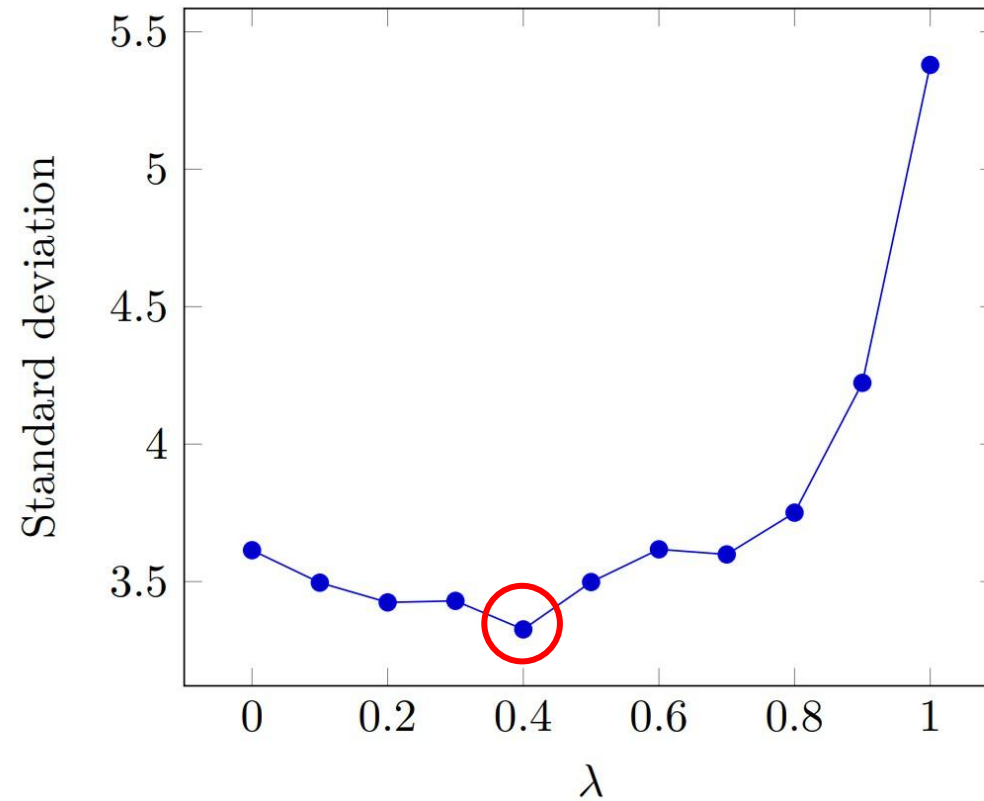
* $(1 - \lambda)$

* λ

Is the ...? » A ... » Va ... » Ne ...

Teachers	Desired workdays	Assigned workdays	Absolute difference	Relative difference
Max	2	10	8	500%
Sarah	10	10	0	0%
Peter	10	20	10	100%

Determining λ



Problem

Reminder:

Instead of minimization of the standard deviation:

- » *Minimize the average relative deviation of desired to assigned workload*
- » *Minimize the **maximum** relative deviation of desired to assigned workload*

Additional constraints (1)

$$\delta_t^+ + \delta_t^- = \underbrace{\left(\sum_c^C x_{tc} \cdot L_c \right)}_{\text{assigned \# workdays}} - \underbrace{W_t}_{\text{desired \# workdays}} \quad \forall t \in [1, \dots, T]$$

$$\frac{\delta_t^+ + \delta_t^-}{W_t} \leq \Delta \quad \forall t \in [1, \dots, T]$$

$$\Delta, \delta_t^+, \delta_t^- \geq 0 \quad \forall t \in [1, \dots, T]$$

Additional constraints (2)

Unsatisfiable scenarios need to be addressed!

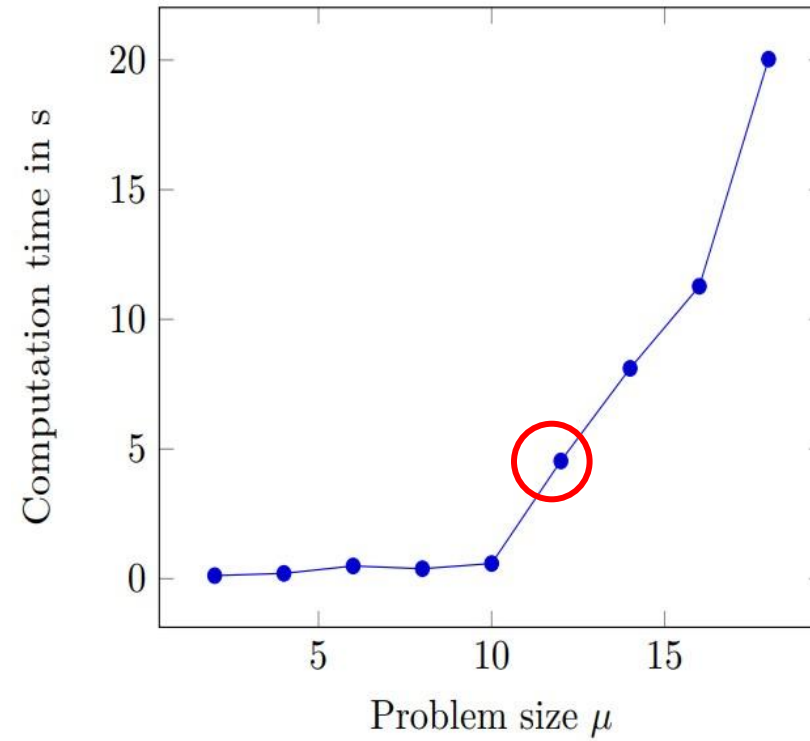
Procedure:

- » Solve a new optimisation problem: what is the maximum possible number of assignments? Call this number y .
- » Add a new constraint: At least y assignments need to be done.

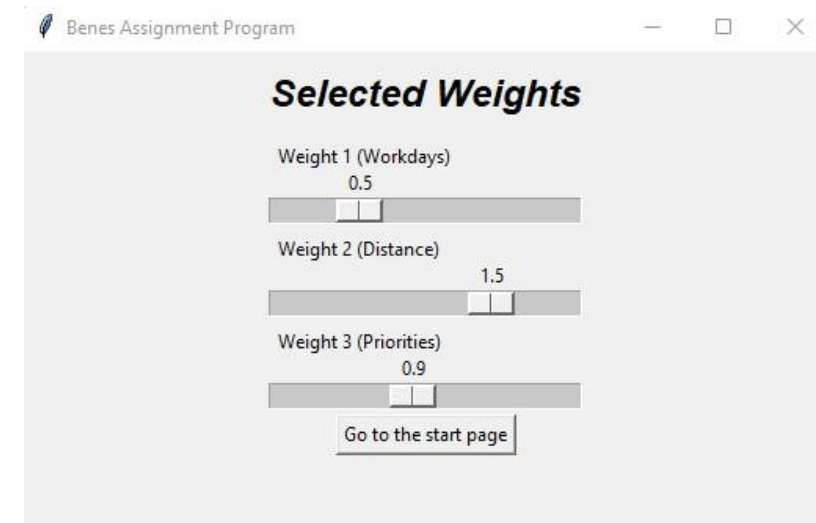
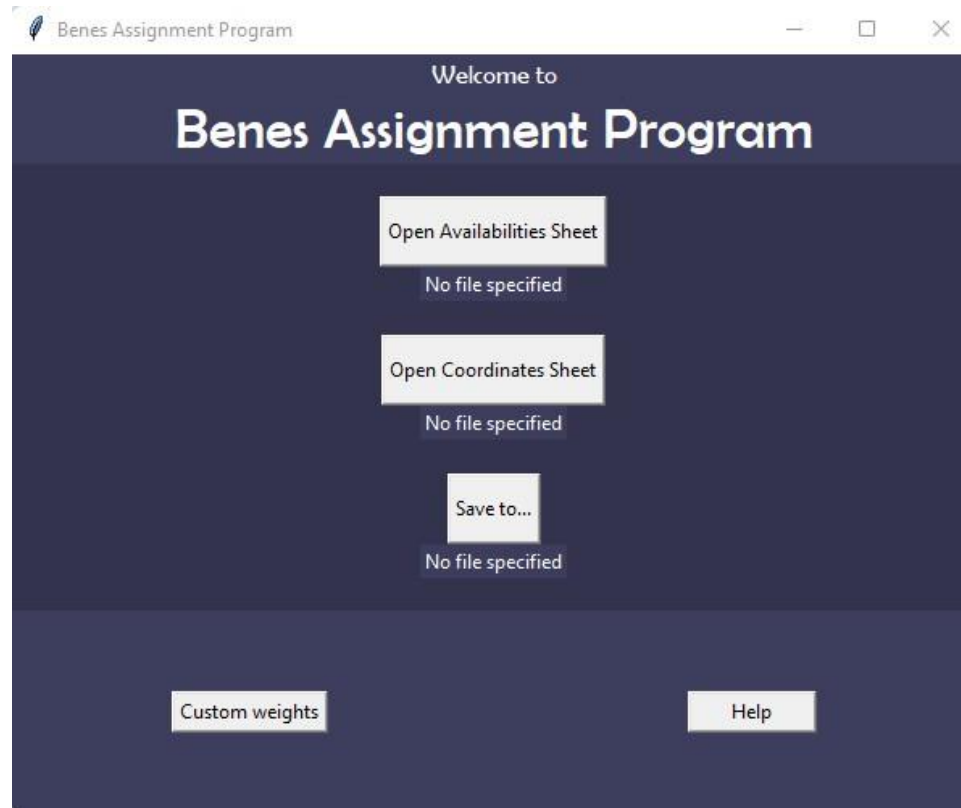
Cost function

$$\begin{aligned} & \frac{\alpha_1}{A} \cdot \sum_{t=1}^T \sum_{c=1}^C x_{tc} \cdot D_{tc} && \left. \vphantom{\sum_{t=1}^T \sum_{c=1}^C} \right\} \text{distance} \\ & + \alpha_2 \left(1 - \frac{\sum_{t=1}^T \sum_{c=1}^C x_{tc} \cdot P_{tc}}{(\sum_{c=1}^C R_c) \cdot \max_{\forall t \forall c} P_{tc}} \right) && \left. \vphantom{\sum_{t=1}^T \sum_{c=1}^C} \right\} \text{preferences} \\ & + \alpha_3 \left(\frac{\lambda}{T} \cdot \left(\sum_{t=1}^T \frac{\delta_t^+ + \delta_t^-}{W_t} \right) + (1 - \lambda) \Delta \right) && \left. \vphantom{\sum_{t=1}^T \sum_{c=1}^C} \right\} \text{workdays} \end{aligned}$$

Computation time



GUI



GUI (2)

Teacher	Workdays	Desired Workdays	Deviation in %
Dante	14,75	8,25	79%
Aleena	0	10,75	-100%
Phoebe	21	21	0%
Luciana	21,25	9,5	124%
Jaylin	51	21,25	140%
Nicholas	19,5	10,5	86%
Meadow	48	24,25	98%
Lane	44	20	120%
Riggs	6,25	24,25	-74%
Morgan	25,75	26,5	-3%
Average deviation		82%	
Maximum deviation		140%	
Travel distance		73280	
Average preference		0,96	
Average absolute deviation		13,4	

Questions?



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